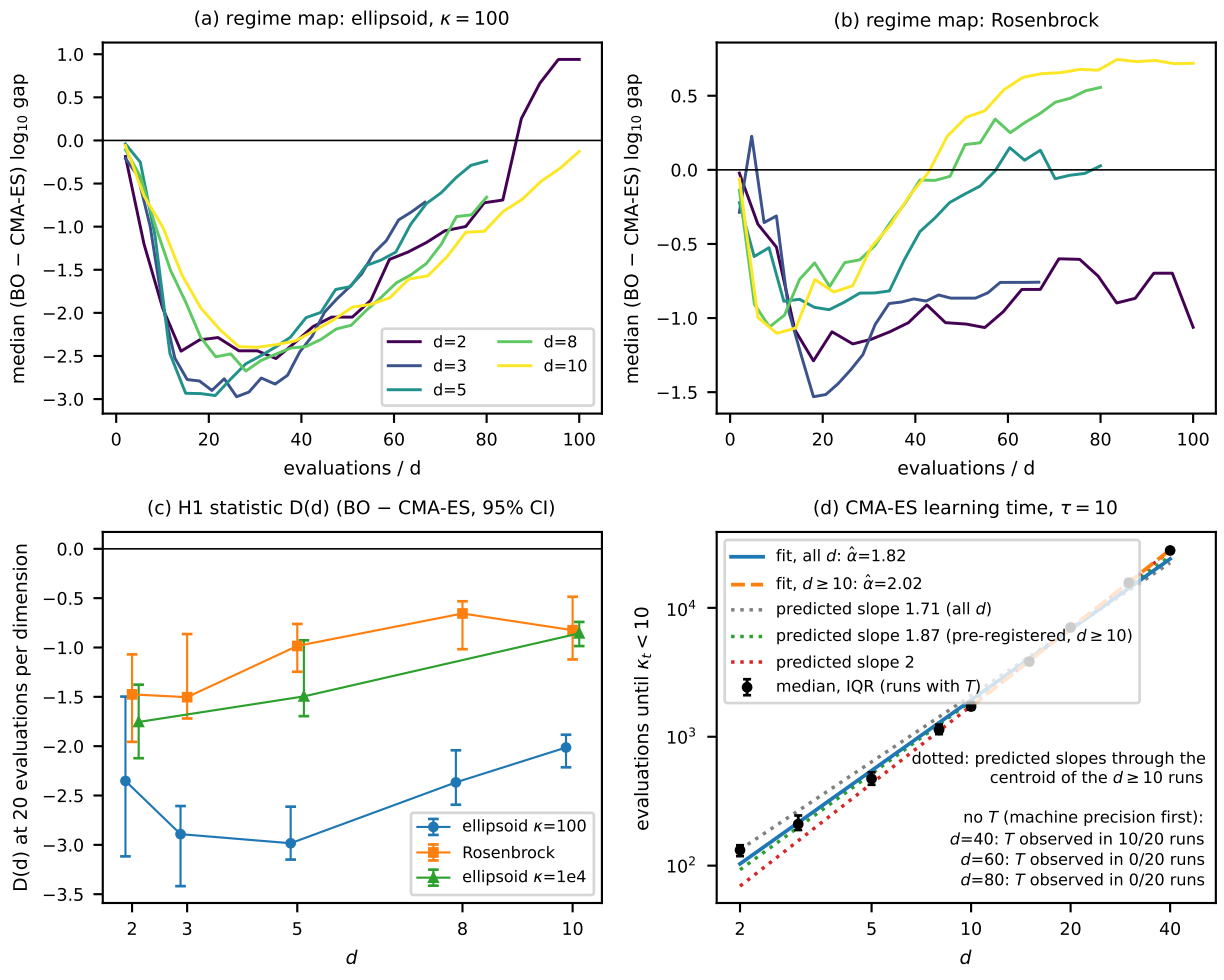
(a) regime map: ellipsoid, $\kappa = 100$
median (BO − CMA-ES) $\log_{10}$ gap
evaluations / d
d=2
d=3
d=5
d=8
d=10
(b) regime map: Rosenbrock
median (BO − CMA-ES) $\log_{10}$ gap
evaluations / d
(c) H1 statistic D(d) (BO − CMA-ES, 95% CI)
D(d) at 20 evaluations per dimension
d
ellipsoid κ=100
Rosenbrock
ellipsoid κ=1e4
(d) CMA-ES learning time, $\tau = 10$
evaluations until $\kappa_t < 10$
d
fit, all d: $\hat{\alpha}$=1.82
fit, $d \geq 10$: $\hat{\alpha}$=2.02
predicted slope 1.71 (all d)
predicted slope 1.87 (pre-registered, $d \geq 10$)
predicted slope 2
median, IQR (runs with T)
dotted: predicted slopes through the centroid of the $d \geq 10$ runs
no T (machine precision first):
d=40: T observed in 10/20 runs
d=60: T observed in 0/20 runs
d=80: T observed in 0/20 runs